

Wealth Intelligence — Technical Blueprint

The intelligence layer between portfolio data and the Relationship Manager. A deterministic engine computes every signal from a private-banking book; a language model only *explains* those signals. This document is written from the code that exists in the repository — every architectural claim maps to a module under `wealth_intelligence/`.

Repository — `wj200/juliusbaer-wi` Interface — Streamlit RM workbench Engine — pure standard-library Python
Signals — 6 detectors · Σ severity² triage Tests — 16 passing Data — synthetic · 20 clients · 5 snapshots

- Python · stdlib core
- Streamlit · workbench
- Claude API · Opus · explanation only
- Docker · container deploy
- unittest · 16 tests
- event_log.csv · authoritative events

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Sections 01–07 describe the system top to bottom; 08–09 are the parts a reviewer scrutinises hardest — the data traps handled correctly and the compliance posture; 10–13 cover how it is verified, shipped, and where it is honest about its own limits.

01 Executive summary & system architecture

The problem. A Relationship Manager (RM) covers twenty private-banking clients across Singapore and Hong Kong. Existing tools are *descriptive* — valuations, allocations, performance — and leave the RM to manually interpret risk, market impact, tax and the next action. The opportunity is an intelligence layer that moves the RM from “*what does the portfolio look like?*” to “*what should I know, and what should I do next?*”

The approach. Everything the system asserts is computed from the source files by a deterministic engine. Each signal it raises carries its severity, a one-line reason an RM could say aloud, the exact numbers behind it, and the source rows as evidence. Only after that does a language model act — and it is handed those findings as *facts to explain*, never as a problem to solve. That boundary is the whole compliance argument: no insight an RM cannot defend, and no number a model invented.

The result. A book-wide triage that answers the challenge's central question — *who does the RM call first, and can she defend the ranking?* — drilling into each client's story with a grounded explanation, and turning every insight into an action the RM accepts, edits or dismisses.

INPUT
12 files

External data · 12 synthetic source files

holdings · transactions · instruments · credit_facilities · mandates · portfolios
clients · commitments · planned_cash_needs · market_context · event_log · rm_notes

join · convert to USD

PHASE 0
model

Book

joined · currency-aware · indexed

Look-through resolver

stock + perpetual + note → one issuer

client_360 → detectors

PHASE 1
engine

Signal engine · deterministic detectors

Collateral

LTV vs call

Concentration

look-through

Mandate

bands · excl.

Liquidity

net of pledge

Income

draw vs yield

Attribution

event_log

Findings → Book triage · rank clients by Σ severity² · every finding carries facts + evidence

----- GOVERNANCE BOUNDARY · deterministic above · the model only explains below -----

explain · facts only

PHASE 2
explain

Grounded explanation · the only LLM in the system

Claude turns findings into a situation read, talking points and honest watch-outs —
handed only computed facts, client profile, RM notes and the cited events. Cached · offline-safe fallback.

render

PHASE 3
workbench

RM workbench · Streamlit

Book triage

Client story + “Why?”

Accept · Edit · Dismiss

decide

Relationship Manager · Priscilla Ong

reviews, decides, and owns the advice

Client advisory conversation

Fig 1 — the pipeline. The book descends one path. The six detectors compute every signal and rank the twenty clients; the dashed line is the governance boundary. Only below it does a language model act. Remove the model and the engine still produces the entire triage — the page just loses its narration.

02 Technology stack

LAYER	CHOICE	WHY
Signal engine	Pure Python standard library	No third-party runtime dependency — runs anywhere a bank permits Python, trivial to audit, nothing to <code>pip install</code> .
Explanation	Claude (Anthropic API), <code>claude-opus-5</code>	Only for turning computed findings into language; adaptive thinking; keyed, cached, and never on the critical path for the numbers.
Workbench	Streamlit	Python-native UI over the engine's JSON — fastest path to the three-zone RM interface.
Tests	<code>unittest</code> (stdlib)	16 regression tests pin the flagship behaviours; no external runner needed.
Deploy	Docker · Streamlit Community Cloud	A container image for a bank-realistic deploy; Community Cloud for a public demo URL.

03 The dataset & ingestion

Twelve synthetic files describe one RM's book: 20 clients, 24 portfolios and 1,015 positions at **five dated snapshots** (Dec 2025 → Aug 2026) spanning a Middle East energy shock, a Strait of Hormuz closure, a June technology drawdown and a rate re-pricing. One snapshot tells you what a portfolio *is*; comparing snapshots tells you what *happened*.

FILE	GRAIN	WHAT IT CARRIES
holdings	1,015 rows	Every position at all five snapshots — the centre of gravity. USD value is authoritative per row.
instruments	62	Price history, <code>underlying_reference</code> (look-through), and the <code>sustainability_excluded / concentration_limit_applies</code> flags.
credit_facilities	5	Lombard / term loans with per-snapshot LTV and margin-call triggers.
mandates	48	Allocation bands, single-position limits, and binding sustainable exclusions.
transactions	393	Trades, income (coupons/dividends/distributions), fees, capital calls, facility activity.
planned_cash_needs	20	Known and expected future liabilities, with certainty and timing.
event_log	16	Authoritative 2026 events with the transmission channels that reached portfolios.
rm_notes	28	The RM's own notes — subjective, and often in tension with the numbers. Where advice lives.

Also: clients, portfolios, commitments, market_context (23 series × 5 dates, incl. ten FX pairs).

04 The signal engine — six detectors

Each detector reads the joined Book and a client, and returns zero or more Findings. Every number is computed from the files; the detectors never guess and are honest where the data is ambiguous. The book-wide engine runs all six across the twenty clients and ranks them by **Σ severity**², so one genuine emergency outranks a pile of minor drift and the ranking is always explainable.

DETECTOR	WHAT IT COMPUTES	FIRES ON
Collateral	Lombard/term LTV vs the margin-call trigger, traced across all five snapshots.	CL-0014 at 69.4% vs a 70% call (SEVERE); CL-0002.
Concentration	Single-name exposure across the whole household, looking through structured products and perpetuals to the name; separates managed from custody.	CL-0014 Golden Harbour ≈29% (stock + perpetual + accumulator); CL-0013 Helios 22.5%.
Mandate	Allocation-band and single-position breaches, plus binding sustainable-exclusion breaches. Custody accounts excluded.	CL-0005 holds excluded energy/palm names in a SUSBAL mandate.
Liquidity	Confirmed near-term needs + uncalled commitments vs what is genuinely sellable — netting pledged collateral , flagging gates and FX mismatch.	CL-0014's "free" liquidity halves once the loan is netted; CL-0017's gated sleeve.
Income	Recurring drawdown vs income the portfolio actually pays. Two branches: a real shortfall, or a retiree drawing on impaired long-dated capital.	CL-0006 covers only 25% of its draw; CL-0012 lives off bonds due 2045 vs a ~2038 horizon.
Attribution	Largest USD moves since baseline, tied to the authoritative <code>event_log</code> by transmission channel.	Tech drawdown → "collateralised lending" → CL-0002/CL-0014 collateral.

05 The look-through resolver

A structured product's asset class tells you what it is *called*; its `underlying_reference` tells you what you are *exposed to*. The resolver maps a holding to the single issuer it really carries risk to, so concentration can be measured the way a risk officer would — counting a stock, its perpetual and an accumulator on it as *one* bet.

- Direct** — the holding *is* the name (a stock, that issuer's perpetual, or a single-underlying note) → one exposure.

- **Basket** — a worst-of / basket note → one exposure per named constituent, flagged so it is not silently merged into the headline single-name number.
- Diversified funds, sovereigns, deposits and gold → no single-name exposure.

The mapping is an **explicit, auditable alias table** — every entry can be pointed at the source field that justifies it. That is the governance posture: no free-association about what a note "probably" references.

Worked example — CL-0014, Lau Chi Ming. Golden Harbour appears three ways: the stock, a 5.25% perpetual, and a 12-month accumulator. Individually each looks modest; rolled up to the name they are $\approx 29\%$ of the managed book — and the accumulator sits in "double-up below strike" territory while his Lombard LTV is 0.6 points from a margin call.

06 Grounded explanation

The engine decides *what is true*. This layer turns a client's Findings into language the RM can use: a situation summary, two to four talking points / actions to consider, and an honest list of what to check. The model is handed **only** the computed facts, the client's profile and objectives, the RM notes, and the cited event rows — and is instructed to reason strictly from them.

The prompt contract

- Reason only from the supplied facts, findings, notes and events. Never introduce a number, holding or world event that is not in the input.
- Every claim must be defensible to the client in a meeting — prefer "we should check X" over a confident assertion the data does not support.
- The RM remains responsible: suggest actions "to consider", never instructions.
- Return a strict JSON object: `situation`, `talking_points`, `watch_outs`.

Robustness

Every explanation is **cached to disk**, keyed by a hash of its exact inputs, so the hero clients can be pre-generated once and the demo runs with **no network**. If there is no API key or any error, the layer falls back to a deterministic explanation assembled from the Findings themselves — the app always works.

Model: `claude-opus-5` with adaptive thinking. The key is read from the environment (or Streamlit Secrets) and never committed. A refusal or failure degrades to the deterministic path.

07 The RM workbench

A single Streamlit surface in three zones, matching *signal* → *understanding* → *decision*:

- **Book** — the triaged book in the sidebar: who to call first, each with a severity chip and a one-line reason.
- **Client** — the selected client's grounded explanation, plus a look-through exposure snapshot.
- **Signals** — every finding with a **"Why?"** expander showing the exact facts and source rows, and **Accept / Edit / Dismiss** so the RM stays in control.

It opens on the highest-priority client (Lau Chi Ming) so the first frame already shows what the system does.

08 Data-handling decisions

The dataset sets traps; handling them correctly is much of the assessment.

DECISION	RATIONALE	WHAT IT COSTS
Value from <code>market_value_usd</code> , never the base-currency AUM columns	Avoids reading a HKD→USD conversion as a loss — Lau's book "falls" 206m→26m only on paper.	None — it is the correct column.
LTV read as <code>drawn</code> / <code>lending_value</code>	Lending value is already advance-rate haircut, per the data dictionary — not raw market value.	None.
Custody separated from managed	A client-directed legacy stake is a concentration to discuss, not a mandate breach to correct.	More logic per finding.
Encumbered collateral netted from "sellable" liquidity	Pledged assets are a prior claim; a book that looks liquid can be anything but.	Requires joining facilities to holdings.
Income annualised from observed quarters, with the estimate labelled	Honest about a run-rate rather than asserting a precise annual figure.	A caveat the RM must read.

09 Trust, governance & compliance

AI-driven advisory must preserve trust and the central role of the RM. The architecture answers each requirement structurally, not with a disclaimer.

- **Explainability** — every insight is computed by a named detector and carries its facts; the "Why?" expander shows them.

- **Suitability** — findings respect mandate, risk profile and objectives; the model is told to as well.
- **Human oversight** — the RM accepts, edits or dismisses every insight; nothing is auto-actioned.
- **Traceability** — each finding lists its source rows as evidence (e.g. `credit_facilities.csv:CF-0002`).
- **Compliance** — the model never computes a number or free-associates about the world; explanations are grounded in the authoritative event log.
- **Security** — no client data leaves the engine; the API key is read from the environment and never committed; the explanation layer sends only synthetic, already-computed facts.

10 Testing strategy

16 standard-library tests pin the behaviour the demo depends on — run with `python -m unittest discover -s tests:`

- **Loading & FX** — book shape (20/24/1,015/62/393) and USD conversion for HKD and EUR.
- **Collateral** — Lau's margin call is SEVERE and <1pt from the line; a comfortable facility stays quiet.
- **Concentration** — Golden Harbour rolls up ≥ 3 instruments and breaches the limit; a custody legacy stake is *not* a breach.
- **Mandate** — the sustainable-exclusion breach fires for CL-0005.
- **Liquidity** — encumbered collateral is netted (free < gross).
- **Income** — the retiree flags impaired capital (not a false shortfall); a genuine shortfall flags; "Pre-retirement" is not treated as retired.
- **Ranking** — Lau tops the book; all 20 clients analyse without error.

11 Deployment & folder structure

```
juliusbaer-wi/
├─ wealth_intelligence/      # the engine — pure standard library
│   ├── data_model.py        # Phase 0 · joined, currency-aware Book
│   ├── lookthrough.py       # Phase 0 · positions → single issuer
│   ├── findings.py          # severity + Finding types
│   ├── detectors.py          # Phase 1 · six detectors
│   ├── engine.py            # Phase 1 · run + rank the book
│   ├── explainer.py         # Phase 2 · grounded NL (Claude)
│   └─ cli.py / __main__.py  # CLI triage
├─ streamlit_app.py          # Phase 3 · RM workbench
```

```
└─ scripts/pregenerate.py      # warm the explanation cache
└─ tests/test_engine.py       # 16 unittests
└─ data/                      # 12 synthetic source files
└─ docs/                      # architecture.html · blueprint.html · DATA_DICTIONARY.md
└─ requirements.txt · runtime.txt · Dockerfile
└─ .streamlit/config.toml     # navy/gold theme + secrets bridge
```

Run it

- **Local** — `pip install -r requirements-app.txt && streamlit run streamlit_app.py`
- **Docker** — `docker build -t jb-wi . && docker run -p 8080:8080 -e ANTHROPIC_API_KEY=... jb-wi`
- **Streamlit Cloud** — point it at `streamlit_app.py`, add `ANTHROPIC_API_KEY` in Secrets, deploy.

For the demo, warm the cache first: `python scripts/pregenerate.py CL-0014 CL-0012 CL-0017` — then it runs with no network.

12 Roadmap

- **Scenario toggle** — "Middle East de-escalates / worsens", re-pricing exposures from the market series to show forward risk.
- **Meeting pack export** — a one-page prep sheet per client the RM can take into the room.
- **More detectors** — currency-mismatch on future liabilities, tax-lot harvesting across a household, KYC-review-due surfacing.
- **Right-sizing for scale** — the stdlib model holds twenty clients in memory; a real book moves to a columnar store behind the same detector interface.

13 Risks, limitations & caveats

- **Income is a run-rate estimate.** Annualised from the observed quarters of transactions; the detector says so rather than asserting a precise figure.
- **Look-through is a curated map.** It is auditable but must be maintained as the instrument universe changes; an unrecognised note contributes no single-name exposure rather than a wrong one.
- **Private-market valuations lag.** Quarterly-reported funds are a quarter behind by design — a feature of the asset class, surfaced honestly.
- **The engine finds only what it encodes.** The model cannot surface a signal the detectors do not compute — a deliberate trade for auditability over open-ended inference.
- **Synthetic data throughout.** No real client, instrument or identifier is present; the habit of treating it as real is part of the exercise.

